

FAST-NU Spring 2026

Subject: ML for Robotics (CS-4090)

Instructor: Dr. Maryam Iqbal

Marks (20)

Assignment# 01 (CLO-1)

Class: BCS 8 A

Date: 16th February 2026

Due Date: 22nd February 2026

Shuttle Bot: The Future of Autonomous Sports Robotics

Introduction

In this assignment, your team will design and develop an autonomous robot capable of detecting, collecting, and placing shuttlecocks at a predefined location. The robot will utilize machine learning (ML) algorithms for shuttlecock detection and control techniques for maneuvering, collection, and precise placement. This project will combine your knowledge of robotics, control systems, and machine learning to build a practical solution for automating a common task in sports environments.

Objective:

The objective of this assignment is to challenge your team to design, construct, and program an autonomous robot that can:

- Detect shuttlecocks on the ground using a machine learning-based vision system.
- Navigate autonomously within a specified area, avoiding obstacles.
- Pick up the shuttlecock using a mechanical arm or collection mechanism.
- Transport the shuttlecock to a designated collection area and place it accurately.
- Perform all tasks efficiently while maintaining stability and precision.



Automate, Navigate, Dominate!

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Instructions & Deliverables

1. Form a team of 3-4 students.
2. Select the components required for the shuttle bot.
3. Conduct a financial analysis and present an **estimate** of the total cost of the shuttle-bot.
4. **Prepare** a report on the design of the shuttle bot. The report should include the following sections:
 - Introduction
 - Materials and Methods
 - Circuit Diagram
 - Financial Analysis

Grading Rubrics

The **assignment** will be graded based on the following criteria:

1. **Design and Innovation** (Marks=5)
 - Creativity in the design of the robot's structure and collection mechanism.
 - Use of cutting-edge ML and control techniques for shuttlecock detection and navigation.
2. **Component Specifications** (Marks=5)
 - List of components required for optimal performance.
 - Specifications of all components
3. **Financial Analysis:** (Marks=5)
 - A detailed financial breakdown of the project, including the cost of materials, components, and any additional expenses incurred during the design and construction process.
4. **Teamwork** (Marks=5)
 - Effective collaboration within the team.
 - Leadership and communication skills of the captain.

You can check out this video that demonstrates an autonomous robot picking up shuttlecocks in action. It shows how the robot detects, collects, and transports shuttlecocks autonomously perfect for inspiration and understanding the mechanisms involved in your own project.

Here's the link to the video: [Autonomous Robot Picking Shuttlecocks](#)

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